ATOMBERG · ATOMQUEST 2026

The Fan That Adapts

A ceiling fan with a **Body** that reshapes with heat, a **Reflex** tuned to climate, and a **Brain** that learns you.

Approach & solution deck · Research-backed · 13 sections

Team: Girish N & teammates · October 2026



three blades · three layers of adaptation

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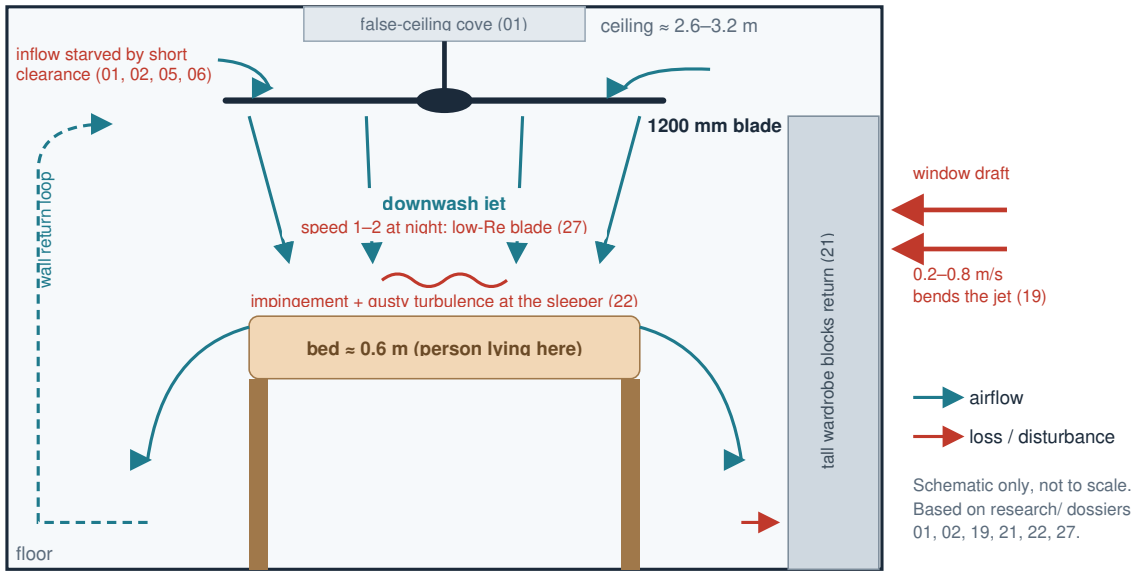
Source files: README · 3 approach reports · 28 dossiers · synthesis (links work once the repo is pushed)

THE PROBLEM

Fans are rated for a room nobody sleeps in

Where a ceiling fan's air goes in a real Indian bedroom (cross-section, schematic)

Ideal fan test = open chamber, max speed. Real use = low speed, bed under fan, walls, furniture, drafts. Numbers = condition IDs.



DESIGNED & RATED

Max speed

open test chamber · one number (air delivery ÷ watts)

VS

ACTUALLY USED

Speed 1–2

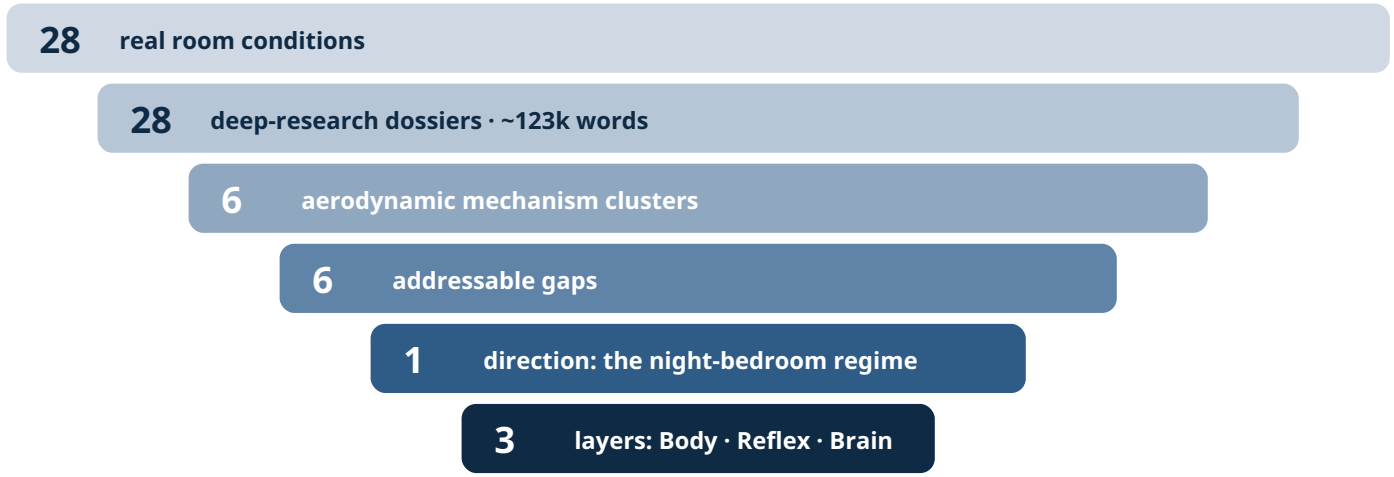
at night · over a bed · real room · heat & humidity changing hour by hour

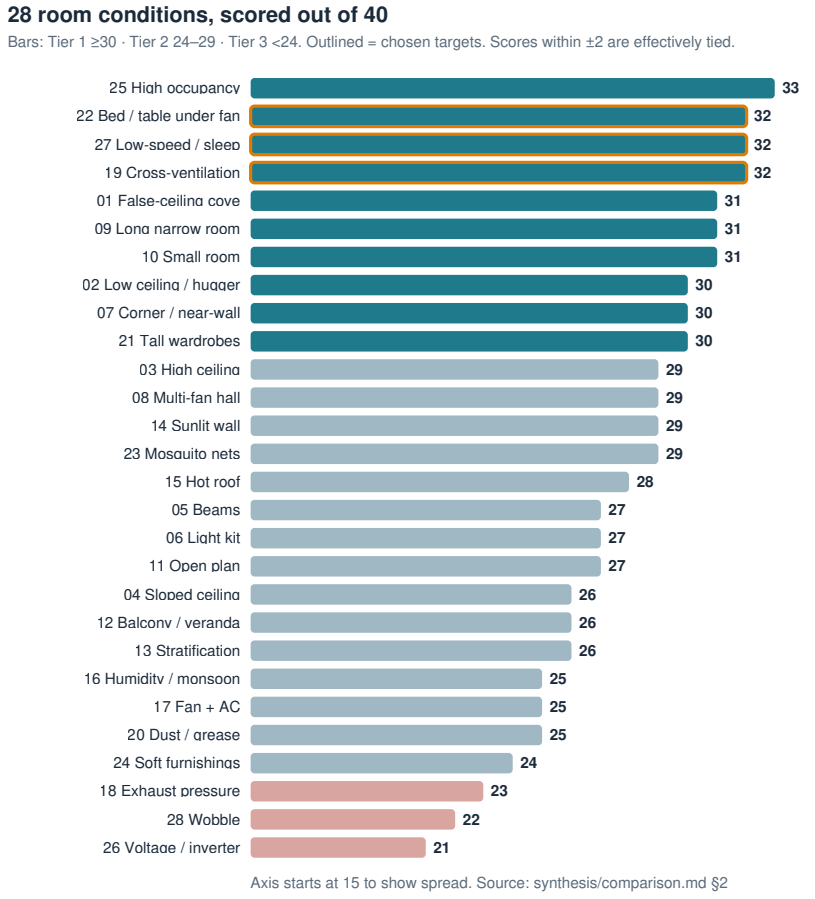
Today's fan behaves the same in all of these situations. **It doesn't sense, reshape or learn.**

HOW DEEP WE WENT

From 28 room conditions to one direction

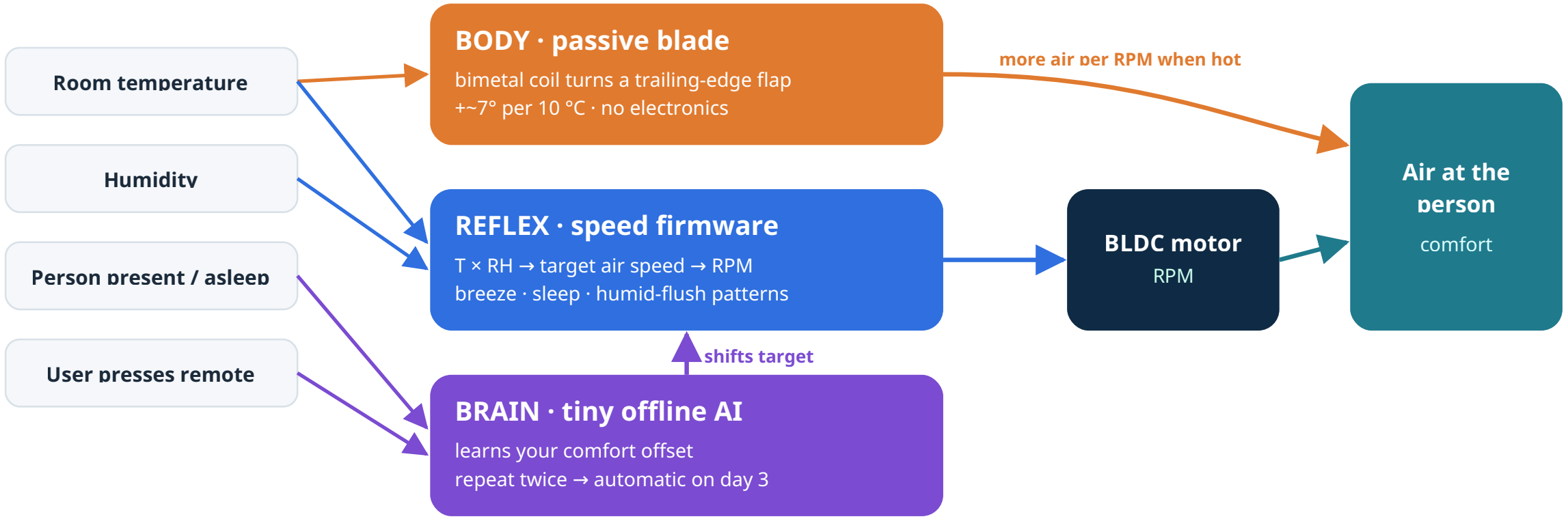
28	~123k	~600	650+	3
room conditions researched	words of dossiers	sources cited	claims flagged UNVERIFIED (honesty)	approach reports · ~24k words





OUR APPROACH

One fan, three layers of adaptation



Body

Works with **no power to electronics, no Wi-Fi, no app**. Most robust, most patentable.

Reflex

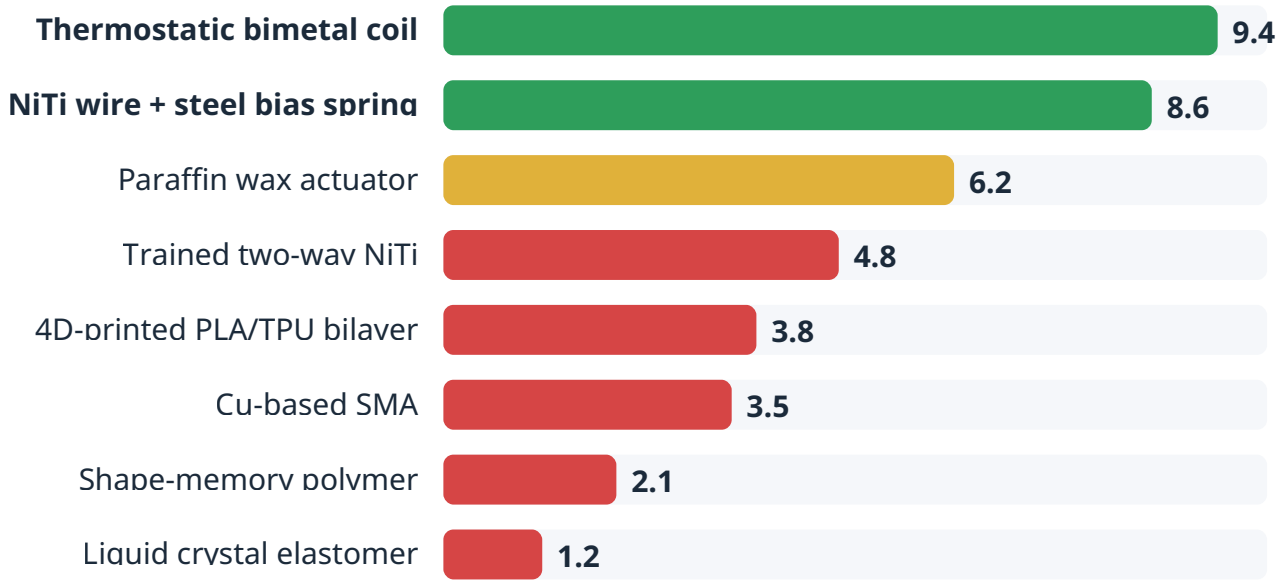
Turns comfort science (ASHRAE 55, Indian IMAC model) into motor behaviour, using hardware the fan **already has**.

Brain

Learns what physics can't predict: **this person** likes it a bit cooler or warmer.

LAYER 1 · BODY

Two shapes, two temperatures: which material?



Feasibility score /10 (from the morphing report)

Key insight: a "true two-way" smart material is the wrong pick. Trained two-way NiTi loses **30–60%** of its motion over time.

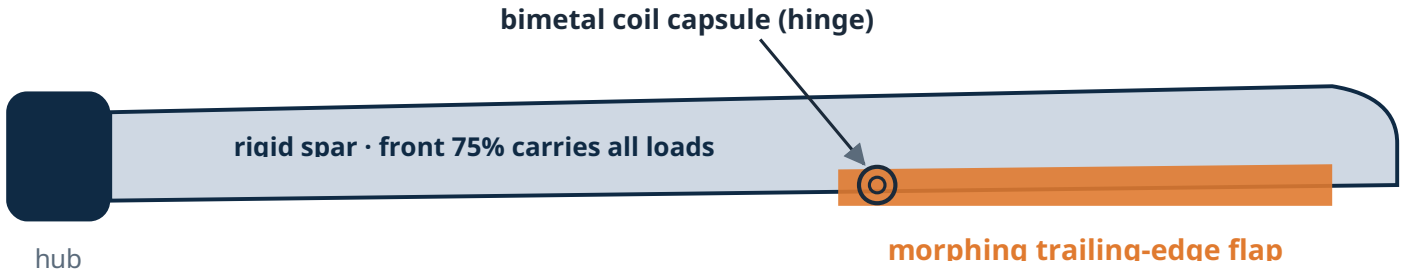
Bimetal = two-way by nature

One-way NiTi + spring = reliable two-way

- **Bimetal:** zero hysteresis, no fatigue, ₹800–1,500/kg, proven for decades in car fan clutches
- **NiTi + spring:** high force, crisp two states; alloy tuned to switch at 27–33 °C
- **Polymers, LCE, 4D prints:** too soft: creep & flutter on a spinning blade

The morphing blade: a flap that reacts to the room

TOP VIEW · one blade (r = 0.10 → 0.60 m)



Centrifugal load (N)



CROSS-SECTION at the flap



COOL 15–24 °C → flat, low pitch
gentle, draft-free circulation (< 0.25 m/s)

WARM 28–38 °C → flap down
more camber · pitch 10° → 13° · ≈ +11–14% air

7.4°
bimetal coil rotation per 10 °C

+11–14%
air from +3° pitch (target 12–18%)

< 1 min
response to room temperature

Concept A (primary)

Helical bimetal coil drives the flap: smooth, proportional, nothing to wear out.

Concept B

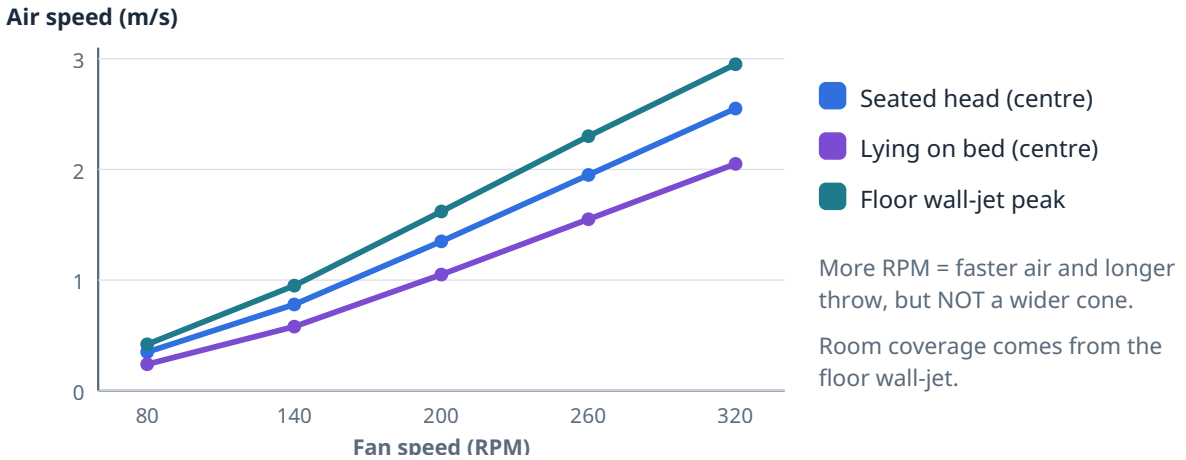
NiTi wire + bias spring rotates the whole blade between two pitch stops; spring = fail-safe.

Must prove

All 3 blades morph identically (tip tracking ≤ 2 mm) · corrosion isolation · travel stops · overspeed test.

LAYER 2 · REFLEX

What fan speed really does to the air



More RPM = faster air and longer throw, but NOT a wider cone.

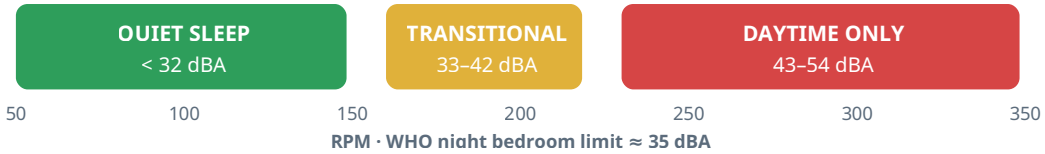
Room coverage comes from the floor wall-jet.

Energy cost of modulating speed (same average RPM)

Steady speed	baseline
Gentle breeze (±15%, 20 s)	+3.8% power
Aggressive (±60, 2 s)	+28.5% power

Claim	Verdict
A special RPM widens the air cone	Disproven
"Natural breeze" fluctuation feels cooler than steady air	Proven
Fast speed cycling saves energy	Disproven
More air always helps in humid heat	Up to ~32 °C / 70% RH

Noise by speed band

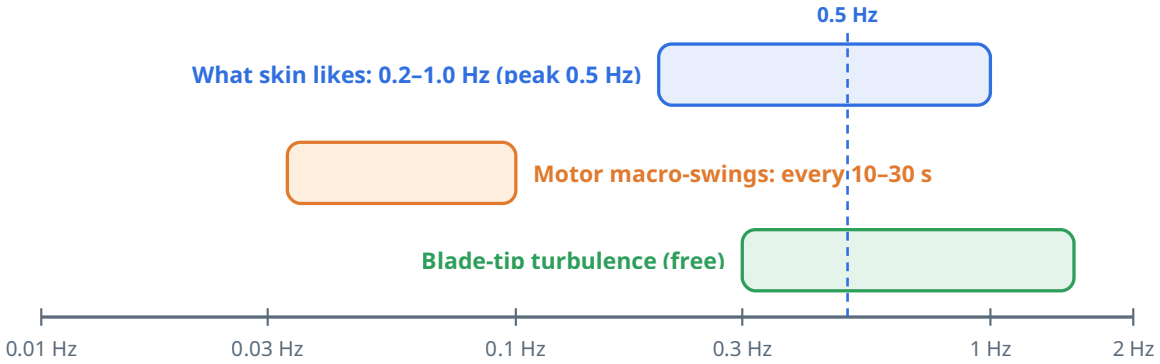


LAYER 2 · REFLEX

Climate map + three breeze modes

	Drier	Humid	Verv humid
24 °C	50–80 40–60%	80–110 70–85%	
26 °C	100–130 30–50%	130–165 60–80%	
28 °C	150–190 30–50%	190–230 60–75%	230–260 >75%
30 °C	200–245 30–50%	245–285 50–70%	285–325 >70%
32 °C	260–300 30–50%	300–335 50–70%	330–350 >70%
34 °C	285–325 30–50%	340–350 >60%	
36 °C+	Cap at 180–200 RPM: beyond this, more air heats you		

Target fan RPM for a 1200 mm BLDC fan · cell = RPM band, small text = RH band



A 1200 mm rotor (inertia ≈ 0.05–0.07 kg·m²) can't swing at 0.5 Hz without heat and noise → slow firmware swings + fast aerodynamic flutter together give the natural-wind feel.

Natural Breeze

Daytime · 26–32 °C



3 slow waves (pink-noise-like) on top of the climate RPM · accel ≤ 12 RPM/s

Sleep Breeze

Night · quiet band



50–150 RPM, < 32 dBA · ±15 RPM over 50 s · accel ≤ 2–3 RPM/s · steps down as room cools

Humid Flush

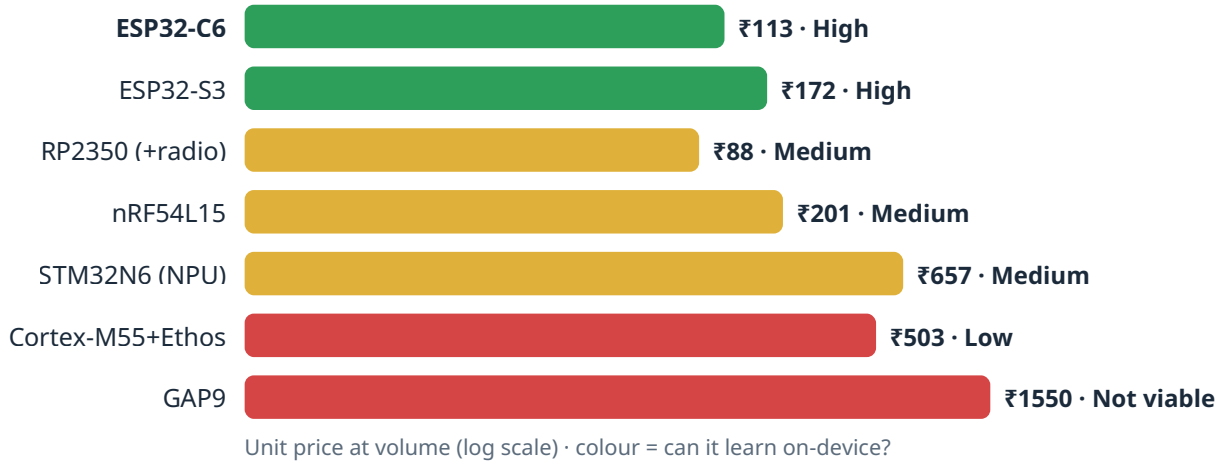
Monsoon / coastal



low base speed + short burst every ~45 s to clear moist air from skin

LAYER 3 · BRAIN

A ₹113 chip, no cloud, no neural network



Sensor	Why
SHT40 temp + humidity	drives the comfort model (critical)
24 GHz mmWave radar	sees a *sleeping* person breathe; PIR misses them
Light sensor	"lights off" = bedtime signal
Motor current / RPM	free: already in the BLDC driver
Remote / app presses	the ground truth it learns from

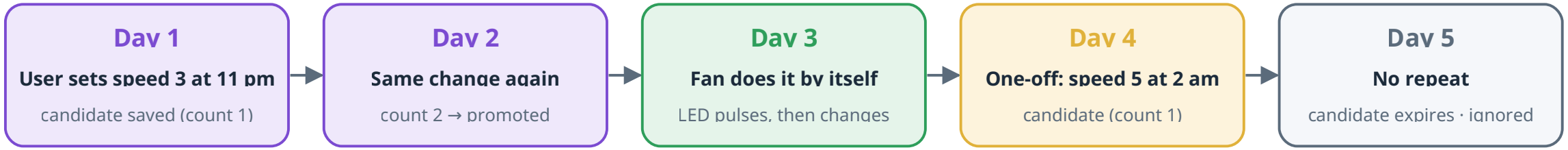
Neural nets on microcontrollers need too much memory, forget habits and wear out flash. A **Bayesian counter + 2-day rule** fits in **< 2 KB RAM**.

Split architecture: the AI never touches motor safety



LAYER 3 · BRAIN

Learns your comfort, not just the clock



Why "comfort offset" beats a schedule: pick speed 4 in a 34 °C heatwave and a clock-based fan still runs speed 4 when the monsoon drops the room to 25 °C. Our Brain stores *"this user likes ~X °C cooler than the model"*, so it follows the weather. It also designs around Nest's schedule-learning patents.

Algorithm	
Habit table + Bayesian update + 2-day counter	Chosen
Moving average	one override shifts it
Bandit (LinUCB)	"explores" at night
Hoeffding tree / tiny RNN	months of data / forgets

Cold start
2 days observing, then a safe comfort curve

Two people
disagreement → lean to lower speed

Seasons
old habits fade; physics takes over

Power cuts
daily save + 12 ms brownout save

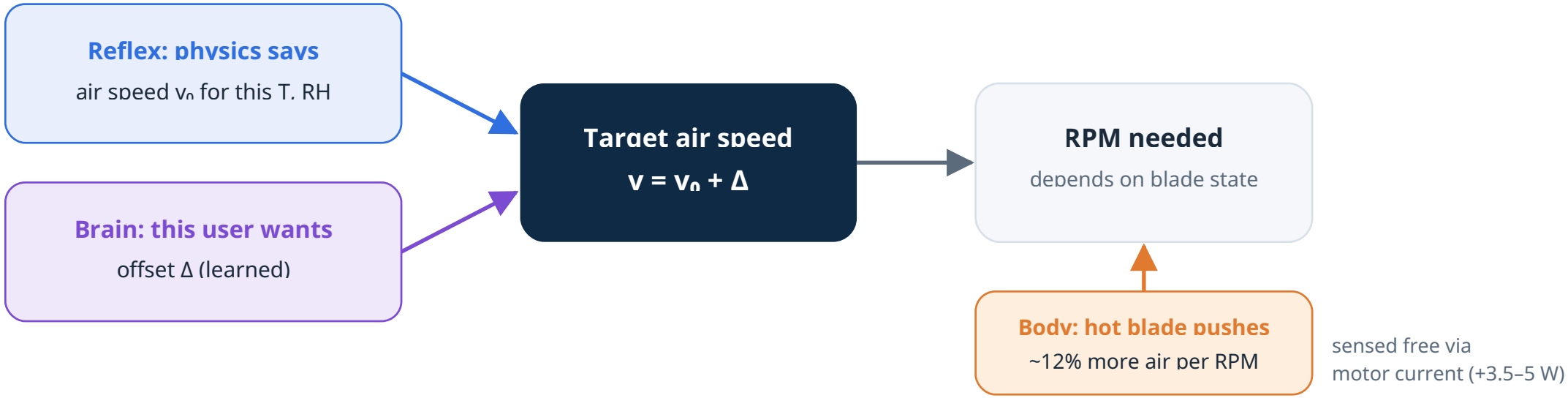
Privacy
100% local · DPDP-friendly · 10 s wipe

Trust
LED pulse first · manual wins in 50 ms

7–10 days
to near-zero manual changes (30-day simulation first)

OUR SYNTHESIS

How the three layers work as one fan



1 · One shared target
The motor aims at *air speed at the person*, not a speed step.

2 · Don't count heat twice
The hot blade already adds ~12% air; firmware reads blade state from motor current so it doesn't overshoot.

3 · Body is the safety net
Electronics off or a plain wall regulator? The blade still adapts.

4 · Flap-made gusts
Using the flap to create micro-gusts at steady RPM is listed as **unstudied**: potentially our most novel claim.

5 · Sleep is where all three meet: quiet band & slow ramps (Reflex) + learned bedtime offset (Brain) + blade flattening as the room cools (Body). This is the night-bedroom regime our phase-1 research ranked #1.

GAPS → SOLUTION

Every gap we found has an owner

	Body	Reflex	Brain
G1 Ratings measure only max speed			
G2 Blades tuned for top speed	●	●	
G3 One fixed blade, two regimes	●		
G4 Sleep modes only slow down	○	●	
G5 Fan doesn't know its room		●	●

● main answer ○ supports · G1 is answered by our own test rig (low speed, at the bed, hot vs cool)

> Phase 1 (28 dossiers) showed **why** fans underperform. Phase 2 (3 reports) shows **how** each gap gets closed, and the test rig produces data **nobody has published**: low-speed, at-the-bed, hot-vs-cool performance.

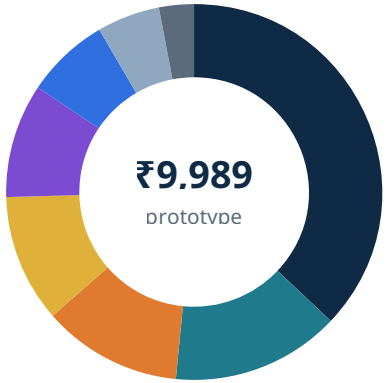
Patent picture	
Passively morphing ceiling-fan blade	no active claim found*
1/f breeze (Panasonic)	expired
SET comfort control (BAF US 9,103,346)	design around
Schedule learning (Nest US 8,630,742)	design around

*to be confirmed with our own patent search

BUILD NEXT

Prototype plan & budget

1. **Static chamber** · 15 → 45 °C box · laser-on-mirror deflection
   proves: morph angle & hysteresis
2. **Spin test** · 50–350 RPM · strobe tip tracking · wattmeter
   proves: balance & power
3. **Airflow test** · vane anemometer grid 1.5 m below · 20 °C vs 35 °C
   proves: +12–18% air claim
4. **Breeze modes** · anemometer at bed height · sound meter
   proves: comfort · < 32 dBA · energy
5. **Habit engine** · ESP32-C6 · 30-day synthetic log → live
   proves: 2-day rule · weather shift



Baseline Atomberg Renesa fan	₹3.699
Tachometer + anemometer	₹1.450
Bimetal strip TB 2011 (1 m)	₹1.200
PETG/ABS filament	₹1.100
NiTi wire (2 m)	₹980
Bearings ×6	₹720
Carbon hinge rods	₹540
Springs ×6	₹300

Morphing-blade BOM from the report (Indian sourcing). Electronics for Reflex + Brain (ESP32-C6 board, SHT40, LD2410 radar) ≈ ₹1,500–2,000 extra (our estimate).

STATUS

Where we are & what we verify next

WE ARE HERE → building prototypes

✓	✓	✓	4	5	6
Problem research	**Synthesis**	**Approach research**	**Morphing blade**	**Firmware + AI**	**Demo + pitch**
28 dossiers	clusters · gaps	3 reports	Concept A → B	modes · habit engine	integrated fan

Verify before the pitch

- ☐ Patent white space for morphing ceiling-fan blades (own search + wind-turbine / propeller prior art)
- ☐ Every patent number in the 3 reports, especially the Indian entries
- ☐ +12–18% air and +3.5–5 W, measured ourselves
- ☐ T×RH → RPM table and 0.2–1.0 Hz band vs original Tsinghua / CBE papers
- ☐ Bimetal flexivity & price from Shivalik datasheet
- ☐ Atomberg sleep-mode step (90 min vs 2 h)
- ☐ IS 374:2019 clause: is low-speed testing really absent?

Honesty note: the approach reports are AI deep-research output. Treat their numbers as **hypotheses to measure**, not facts, until the test rig confirms them. That is exactly what the prototype plan is for.

"Designed for the test chamber" →
"designed for the person under the fan."